

HAO YU

<http://cs-people.bu.edu/haoyu/>

Department of Computer Science, Boston University

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RESEARCH INTERESTS

I am intereseted in computer vision and machine learning.

EDUCATION

Boston University

Sep 2019 - Present

PhD Student in Computer Science

Advisor: Prof. Margrit Betke

Zhejiang University

Sep 2015 - Jun 2019

B.E. in Computer Science and Technology

B.A. in English Language and Literature

Chu Kochen Honors College

Overall GPA: 3.95 / 4.0

Advisor: Prof. Jian Wu

National University of Singapore

Jan 2019 - May 2019

Research Student

Advisor: Prof. Terence Sim

University of California, Davis

July 2018 - Sep 2018

Research Student

Advisor: Prof Yong Jae Lee

PUBLICATIONS

[1] Krishna Kumar Singh, **Hao Yu**, Aron Sarmasi, Gautam Pradeep, Yong Jae Lee, Hide-and-Seek: A Data Augmentation Technique for Weakly-Supervised Localization and Beyond, arXiv preprint arXiv:1811.02545, 2018.

RESEARCH EXPERIENCE

Fake Face Video Detection Using Talking Profile

January 2019 - April 2019

Under the supervision of Prof. Terence Sim

National University of Singapore

- Developed a detection algorithm for recognizing fake face videos generated by face swapping.
- Used talking profile which consists of multiple types of facial and body motions to identify fake face videos.
- Trained a SVM classifier of talking profiles for each subject using real videos and test using fake videos.

Hide-and-Seek: A Data Augmentation Technique

Jul 2018 - Sep 2018

Under the supervision of Prof. Yong Jae Lee

University of California, Davis

- Applied Hide-and-Seek on improving performance of CNN models of Emotion Recognition, Age Estimation, Gender Estimation and Person Re-identification. The key idea of Hide-and-Seek is to randomly hide patches in a training image, which increases the variety of dataset while preserves spatial alignment.
- Achieved superior performance compared to previous methods for all tasks on the corresponding benchmarks.

- Summarized my work and coauthored the paper Hide-and-Seek: A Data Augmentation Technique for Weakly-Supervised Localization and Beyond.

Recognition of Invasive Cervical Cancer Based on CNN

May 2017 - May 2018

Under the supervision of Prof. Jian Wu

Zhejiang University

- Developed a visual recognition system of Colposcopy images for auxiliary diagnosis of invasive cervical cancer.
- Applied SE-ResNet on classification of colposcopy images.
- Obtained 90.86% sensitivity and 73.94% specificity on a dataset of 8556 images, which reached the level of qualified doctors.

PROFESSIONAL ACTIVITIES

Reviewer for IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)

2019

HONORS AND AWARDS

- Meritorious Winner of COMAP's Interdisciplinary Contest In Modeling (ICM) *2017, 2018*
- First-Class Scholarship for Excellence in Research and Innovation in ZJU *2017, 2018*
- Second-Class Scholarship for Outstanding Merits in ZJU *2016, 2018*

RELEVANT COURSEWORK

- **Graduate Coursework:** Machine Learning, Big Data Systems for Data Science.
- **Undergraduate Coursework:** Computer Vision, Artificial Intelligence, Computer Graphics, Information Retrieval.

TECHNICAL STRENGTHS

Computer Languages

Python, C/C++, Java, Matlab, LaTeX

Deep Learning Frameworks

Tensorflow, PyTorch, Caffe, Keras

Toolkits

OpenCV, OpenGL